

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
2	(a)		Angle when arc [length] equals radius Accept about 57.3° or angle when $2\pi = 360^\circ$ or cycle / circle	1			1		
	(b)	(i)	Use of $T = \frac{1}{f}$ (1) Answer = 1.67 [s] (1) Accept $\frac{5}{3}$ or 1.66 or 1. $\dot{6}$ or 1.7 [s] Don't accept 1.6 [s]	1	1		2	1	
		(ii)	Substitution into $\omega = \frac{2\pi}{T}$ or $2\pi f$ or and $v = \omega r$ (1) ecf on T or f $v = 10.6$ [m s ⁻¹] (1) (Accept 10.5 m s ⁻¹ if 1.67 s used)		2		2	2	
	(c)		$N = \frac{mv^2}{r}$ or $mr\omega^2$ or implied (1) $N = \frac{66.2 \times (10.6)^2}{2.8} = [2\ 634]$ [N] (1) ecf on v and ω accept approximately 2 657 [N] $F = 66.2 \times 9.81 = [649.4]$ [N] (1) Vertical forces are balanced or equivalent e.g. $F = W$ (1)	1 1	1 1		4	2	
	(d)	(i)	$650 \leq \text{or} = \text{or} < \mu \times 2\ 600$ (1) So $\mu > \text{or} = 0.24$ or 0.25 (1) Alternative: $2\ 600 \times 0.25$ (1) $= 650$ (1)	1	1		2	2	
		(ii)	Friction = 650 [N] or implied (1) $\frac{650}{0.45} = 1\ 444$ [N] (1) Equating to centripetal (1) $\omega = 2.8$ [rad s ⁻¹] (1) Answer of 2.51 [rad s ⁻¹] award 1 mark only		1	1 1 1	4	2	
			Question 2 total	5	7	3	15	9	0